Niagara Generation Modeling Update

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Background

- In February and May of this year, NYISO described for stakeholders a modeling enhancement of the Niagara Power Project in the Real-Time and Day-Ahead Markets planned for the end of 2018. Please see the materials from those meetings:
  - http://www.nyiso.com/public/webdocs/markets_operations/committees/bic_miwg/meeting_materials/2018-02-21/Enhancements%20to%20the%20Niagara%20Model_MIWG2018.pdf
  - http://www.nyiso.com/public/webdocs/markets_operations/committees/bic_icapwg/meeting_materials/2018-05-09/Enhancements%20to%20the%20Niagara%20Model%20vFINAL.pdf
- In response to Market Participant requests for additional details of this enhancement, NYISO is returning to today's MIWG.

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Review of the Niagara plant modeling

(previously presented at various MIWG meetings)

- **The Niagara Power Plant is comprised of 25 individual generating units, divided into three distinct points of injection onto the bulk power system:**
  - Niagara 230kV Bus:
    - 12 Lewiston Pump Generating units (480MW total)
    - 6 Main Generating units (1290MW total)
  - Niagara 115kV East Bus
    - 4 Main Generating units (860MW total)
  - Niagara 115kV West Bus
    - 3 Main Generating units (645MW total)

- **Since 2016, the NYISO's energy market software assumes that all incremental injections from the Niagara plant will be distributed across the three distinct Niagara buses according to a static ratio over the forward time horizon of the RTD/RTC/SCUC models.**
  - Plant distribution assumptions for the Real-Time Market are based on actual telemetered output from the individual units observed at the time each RTD/RTC execution initializes.
  - Plant distribution for the Day-Ahead Market is generally a single static value applied to all hours determined based on the previous like day observations.

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What is being enhanced in the model

(previously presented on February 21, 2018)

- NYISO is working on a modeling enhancement that provides the Real-Time and Day-Ahead Markets greater freedom in the assumptions about the distribution of Niagara injections across its three bus locations to address transmission constraints.
  - Rather than employ a static distribution assumption that is developed at the time of initialization of RTD/RTC/SCUC and used for all forward time steps, the market software will instead establish its own distribution assumption of the Niagara plant's injections for each time step based on the impacts of each bus on transmission constraints.
- This model enhancement enables RTD/RTC/SCUC to more effectively address transmission constraints consistent with the available generation capabilities.
  - It will make the dispatch of the plant be predictive of the actions taken, rather than reactive.
- This flexibility is also an important prerequisite to securing the Niagara 115 kV transmission constraints in the market models given the significant (and different) impacts on those constraints of output at the three bus locations.

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2017 State of the Market report

"Enhanced modeling of the Niagara Plant would significantly reduce the costs of managing 115 and 230 kV congestion in the West Zone. The plant consists of 7 generating units on the 115 kV network and 18 generating units on the 230 kV network, and output can be shifted among these generators to manage congestion on both networks and make more of the plant's output deliverable to consumers. However, the NYISO currently represents all 25 units as a single facility in the market models, which prevents the NYISO from shifting generation among these units to manage congestion and increase output from the plant. Instead, the market scheduling systems turn the entire facility up or down without distinguishing between generators that relieve congestion and generators that exacerbate congestion, which increases the costs of congestion management. The NYISO recently recognized this and announced an initiative to improve the modeling of the Niagara generator."

*Page 29 of 2017 State of the Market Report:
http://www.nyiso.com/public/webdocs/markets_operations/documents/Studies_and_Reports/Reports/Market_Monitoring_Unit_Reports/2017/NYISO-2017-SOM-Report-5-07-2018_final.pdf

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The practice in operations

(previously presented on February 21, 2018)

- The market model NYISO used prior to mid-2016 assumed the 2,200 MW would all be injected at the Niagara 230kV bus.
- The market model NYISO has used since 2016 assumes the 2,200 MW would be injected according to the actual plant distribution metered at the at the time of each RTD/RTC initialization.
- The enhanced market model that NYISO is developing will distribute the 2,200 MW in a manner that minimizes congestion on the transmission system. The optimization will have the ability to distribute the schedule similar to how the operators currently request re-distribution at the plant.

Under the three different models, the Operators may request the plant to re-distribute generation if overloads are observed.

If overloads cannot be addressed through re-distribution under the first two models due to individual bus limitations, the entire plant may be operated out-of-merit to preserve reliable operations.

*If overloads cannot be addressed through re-distribution under the enhanced model due to individual bus limitations, operators will be able to represent those limitations to the dispatch, * reducing the frequency that the entire plant needs to be operated out-of-merit.*

** The available dispatch input parameters that can effectuate those limitations are described on the next slide*

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Dispatch input parameters

- The enhanced model will employ the following dispatch input parameters, which can be adjusted by Operators to reflect transient limitations of the plant
  - 115E kV Bus minimum and maximum MW
  - 115W kV Bus minimum and maximum MW
  - 230 kV Bus minimum and maximum MW
  - Total plant minimum and maximum MW
- NYISO expects the maximum MW to be the typical input parameter used.

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Benefits

- In 2017, NYISO manually instructed the Niagara plant to shift output between its generators for approximately 1,200 hours to manage constraints on several 115 kV and 230 kV facilities in the West.
  - Over the years, this procedure has proven to be an effective tool to manage western constraints.
  - The vast majority of the time the plant is able to accommodate these instructions.

- With the enhanced Niagara model, the market systems will now be able to predict the need for this shift, will develop a total plant schedule assuming a least-cost distribution for that total plant schedule, and will deliver the distribution instructions to both the plant and to the NYISO Operators.

- If redistribution of plant output is needed, but the plant is unable to achieve the recommended distribution due to a limitation at one of its buses, the NYISO Operators can restrict the market systems to reflect those limitations by adjusting the relevant dispatch input parameters listed on the previous slide.

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Simulation

- The NYISO ran a number of simulations of the enhanced model using production scenarios. Below is an example of the results of one of those simulations:

West 230kV Active Constraints with Shadow Price			
LIMITING_FACILITY	CONTINGENCY_NAME	Current Niagara Model	Enhanced Niagara Model
PACKARD 230 SAWYER 230 1	BUS:PACKARD BK3 & 61 & 78	$1,175.00	Not binding
PACKARD 230 SAWYER 230 1	PACKARD_-SAWYER___230_77	$1,175.00	Not binding
PACKARD 230 SAWYER 230 1	BUS:PACKARD BK4 & 62 & 77	Not binding	$58.28
		Current Niagara Model	Enhanced Niagara Model
LBMP	LBMP Reference Bus	$38.08	$34.25
	LBMP Niagara Plant (PTID 23760)	($44.55)	$40.76
Distribution	% MWs injected at 115 kV East Bus	26%	32%
	% MWs injected at 115 kV West Bus	26%	24%
	% MWs injected at 230 kV Bus	48%	43%

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Example scenarios using the simulation results:

Scenario	NYISO Operations Action
The plant can achieve the overall plant schedule at the expected distribution.	This is the expected outcome the majority of the time. No action needed.
The plant can achieve the overall plant schedule, but they communicate to NYISO an operating limitation that prevents an increase of the 115 kV East bus injections to the desired distribution percentage.	Similar to an operating limitation communicated to the NYISO for any generator, NYISO Operators will reflect that limitation once the information is received so that it can be picked up in the next dispatch run, In this example, the Operators would put a limit on the 115 kV East bus Maximum MW for the next dispatch run.
The plant communicates to NYISO that it cannot meet the overall plant schedule due to an operating limitation.	Similar to an operating limitation communicated to the NYISO for any generator, NYISO Operators will reflect that limitation once the information is received to be picked up in the next dispatch run, In this example, the Operators would put an overall limit on the plant schedule via an out-of-merit.

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Next steps

- NYISO is currently developing the tools and procedures to support the enhanced market model for Niagara.
- NYISO is targeting December 2018 to have the necessary work completed and ready for production use.

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- Planning the power system for the future
- Providing factual information to policy makers, stakeholders and investors in the power system



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